Two thermal zones with one temperature measurement

Original learning collection – November 2026

Evidence note. Original teaching scenario. Numerical parameters are deliberately chosen for learning. All generated data and curves are synthetic; no physical measurement is asserted.

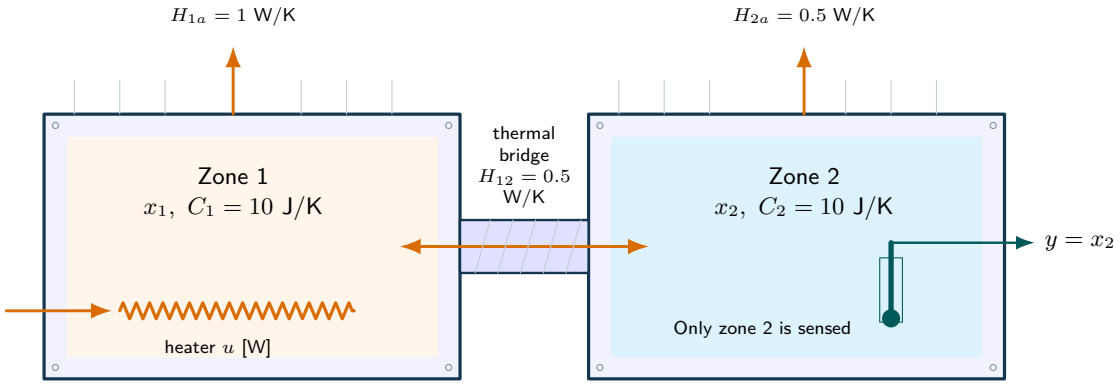
1 Design brief

Two hypothetical thermal zones obey $\dot{x} = Ax + Bu$ and $y = Cx$, with

$$A = \begin{bmatrix} -0.15 & 0.05 \\ 0.05 & -0.10 \end{bmatrix} \text{ s}^{-1}, \quad B = \begin{bmatrix} 0.1 \\ 0 \end{bmatrix} \text{ K/(s W)}, \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}.$$

Check controllability and observability. Choose feedback poles $-0.20, -0.35$, observer poles $-0.8, -1.0$ and a constant reference prefilter. Starting at $x(0) = [0.6, -0.4]^{\mathsf{T}}$ K with zero estimate, regulate zone 2 to a 1 K rise.

2 Physical system



Temperatures are deviations from a fixed ambient temperature

Two energy stores
$C_1 = C_2 = 10$ J/K.
x_1, x_2: ambient-relative temperatures.
Input heat enters zone 1 only.

Coupled balances
$10\dot{x}_1 = u - 1.5x_1 + 0.5x_2$.
$10\dot{x}_2 = 0.5x_1 - x_2$.
All right-hand terms are heat rates.

Sensing and estimation
$y = x_2$ supplies one measured coordinate. An observer reconstructs both states from the applied input and this output.

Two thermal inventories are connected by a conductive bridge. Heating acts in the first zone and sensing in the second. These stated capacities and conductances reproduce the two-state model exactly under the lumped assumptions.

3 What can be controlled and observed?

The controllability matrix $[B\ AB]$ has determinant 0.0005, while the observability matrix $[C; CA]$ for $C = [0\ 1]$ has determinant -0.05. Both are nonsingular. Heat injected into zone 1 can influence zone 2 through coupling, and the measured zone-2 temperature reveals the other state dynamically.

4 Feedback and prefilter

Choose $u = -K\hat{x} + Nr$. Pole placement yields K such that $A - BK$ has eigenvalues $-0.20, -0.35$. For constant references,

$$N = -\left[C(A - BK)^{-1}B\right]^{-1}.$$

This formula gives unit DC reference gain only for the nominal model and no constant disturbance. It is not integral regulation.

5 Observer and separation

Use $\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$ and select L by placing the eigenvalues of $A - LC$ at $-0.8, -1.0$. With error $e = x - \hat{x}$,

$$\begin{bmatrix} \dot{x} \\ \dot{e} \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x \\ e \end{bmatrix} + \begin{bmatrix} BN \\ 0 \end{bmatrix} r.$$

The block triangular matrix shows why controller and observer eigenvalues combine without changing each other in the nominal linear model.

6 Interpretation and limits

Observer convergence can be faster than output regulation. Faster estimation is not free: sensor noise and unmodeled dynamics are amplified by aggressive observer gains. Heating/cooling limits and temperature-dependent heat transfer are omitted; negative incremental heat can require an active cooler or reduction from a positive bias.

7 What to learn from the experiment

State estimation and reference prefiltering solve different problems. A nominal prefilter does not replace integral action against unknown constant disturbances.

8 System sketch

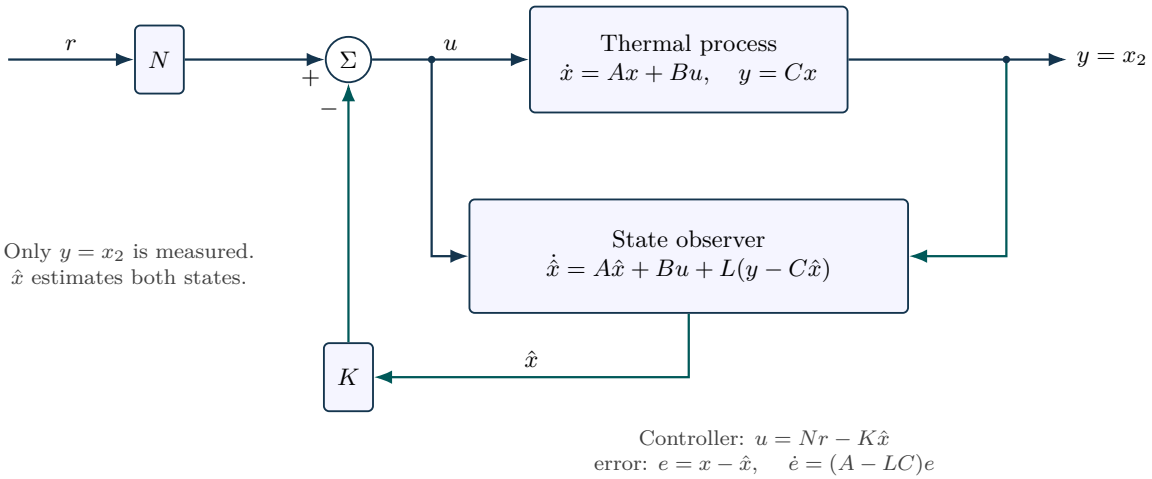
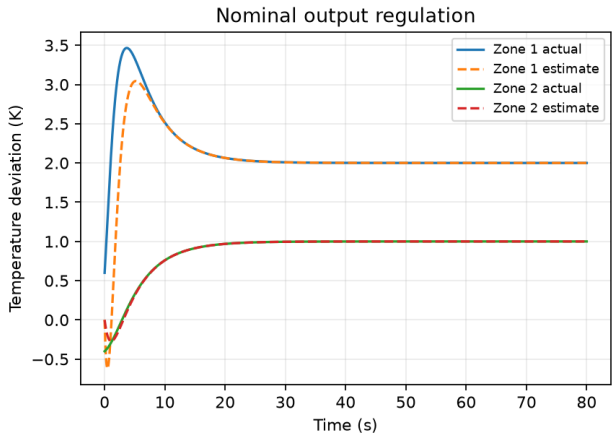
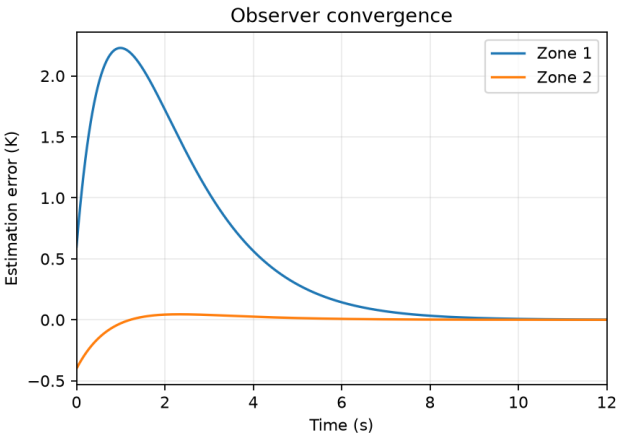


Figure 1. The observer receives both the applied control and measured output. Its estimated vector supplies state feedback; the prefilter sets nominal reference gain.

9 Executed computational evidence

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Feedback K: [[3.  5.5]]
Observer L: [11.1   1.55]
Reference prefilter N=14.00000000
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10 Method references

The equations use standard balance laws and feedback-control methods. Further explanation: https://apmonitor.com/pdc/ and https://fbsbook.org/. Implementation documentation: https://python-control.readthedocs.io/en/0.10.2/. See the repository reference catalog for individual topics and videos.